Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Brainy Barbies
Project / Solution Name	Future Path
Name (Leader)	Law Jing Xuan
Email & Phone (Leader)	lawjingxuan1234@gmail.com (016-2299315)
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Live Prototype URL	https://future-path-ai-lawjingxuan1234.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/jnghan07/SDC-Brainy-Barbies

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

FuturePath uses a frontend-backend architecture and is deployed on Replit. Users interact with the frontend to upload a CV or manually enter their skills and experience. For CV uploads, the backend extracts the text from the PDF and removes obvious personal contact details before sending the relevant content for analysis.

The backend communicates with the Google Gemini API using a securely stored API key. Gemini analyses the candidate's information and returns identified skills, supporting evidence, and relevant skills that have not yet been demonstrated. The frontend then displays these results to the user.

Additional modules, including career matching, skill-health assessment, learning recommendations, SDG mapping, and future scenarios, are designed to extend the analysis. These modules require their respective matching logic, assessment outputs, and supporting datasets to be integrated before all recommendations can be generated dynamically.

The application is hosted on Replit. The current workflow does not require a database for permanently storing uploaded CVs, as CV processing is handled in memory.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Replit Core	Development and deployment for the web application	Replit Terms of Service; not an open-source software licence.
React	Builds the frontend user interface	MIT
TypeScript	Adds static typing to JavaScript code	Apache-2.0

Name & Version	Purpose in Project	License
Node.js	Runs the backed server and API routes	MIT
Qwen3-VL-4B-Instruct	Extracts structured CV information from pdf, png, jpg and jpeg	Apache- 2.0
RunPod Serverless	Hosts the model inference endpoint on GPU	RunPod Terms of Service; model licence is separate
Google Gemini API	Analyses candidate skills and provides career-related insights	Google API terms and applicable service terms; not an open-source licence
Hugging Face	Accesses model resources and supports AI development workflows	Depends on the specific library, model, or dataset used
Google Colab	Runs Python notebooks and model development experiments	Google Colab terms of service
Kaggle	Provides open datasets for model training and demonstrations	https://creativecommons.org/publicdomain/zero/1.0/

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API - RunPod	Hosts the Qwen3-VL-4B-Instruct model on GPU to extract structured candidate information from uploaded CVs.	Sponsor-provided
Gemini API - Google	Analyses users extracted skills and experience to provide career-related insights and recommendations	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
RunPod Serverless - RunPod	GPU hosting and inference for the Qwen3-VL-4B-Instruct model used for CV information extraction.	SDC-sponsored GPU credits
Google Colab – Google	Python notebooks for AI development and training experiments.	Free tier

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-VL-4B-Instruct	Qwen; accessed through RunPod Serverless	Pre-trained model used for CV information extraction.	Apache- 2.0, subject to the model's official license.
Google Gemini	Google Gemini API	Pre-trained model accessed through the API for skills analysis and career-related insights.	Google's applicable API terms and usage policies.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Resume Dataset (2,484 PDF CVs, 24 job categories)	Resume Dataset	Public	https://creativecommons.org/publicdomain/zero/1.0/	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

Yes

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT, Claude	Assistance with code explanations, debugging, model-training guidance, documentation, project planning, and preparation of submission materials.	Major
Replit Agent	Assistance with application development, code generation, debugging, or modifications within Replit.	Major
Google Gemini	Analysing users' skills within the application.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-generated code, documentation, and model outputs to check their correctness and suitability for the project. We tested the CV extraction workflow using sample CV files and checked whether the model returned the expected structured JSON fields. We also tested API responses and validated output formats and error handling. AI-generated suggestions were treated as assistance rather than automatically accepted. Team members reviewed and tested the final implementation, remained responsible for the submitted code and content, and made decisions about what to include in the prototype.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

Yes

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core credits	Used as the primary development environment to build the CV Extraction and Career Recommendation web application.
RunPod	Sponsored GPU credits	Used to run the Qwen3-VL-4B-Instruct model through a RunPod Serverless endpoint for AI-powered CV information extraction.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor-provided resources were important to the development and operation of our prototype. Replit Core supported the development of our web application, while RunPod's sponsored GPU resources enabled us to run the Qwen3-VL-4B-Instruct model for CV information extraction. Although alternative development platforms and GPU inference providers could be used, replacing these resources would require additional setup, testing, and potentially extra costs. Therefore, the resources were highly valuable to delivering our prototype within the hackathon timeframe.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

FuturePath AI processes users' CVs to extract relevant information, such as skills, education, projects, and work experience. Uploaded CVs are processed in memory and are not permanently stored by the application. Before the extracted text is sent to the Google Gemini API for analysis, obvious personal contact details, such as email addresses, phone numbers, and URLs, are removed. The Gemini API key is stored securely on the server and is not exposed to users. Although the application does not permanently store uploaded CVs, some CV content is sent to a third-party AI service for processing. Therefore, users should also avoid uploading sensitive personal information. Nonetheless, we use fictional CVs for demonstration purposes.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

FuturePath AI depends on the Google Gemini API, so service interruptions, rate limits, or API errors may affect its availability. The AI may incorrectly identify skills, misunderstand CV content, or generate inaccurate recommendations. Its analysis may also contain bias and may not fully reflect a candidate's actual abilities. In addition, career recommendations and skill assessments depend on the quality and availability of matching logic, workforce trend data, and learning resources. Outdated or incomplete data may lead to unsuitable recommendations. To reduce these risks, the application displays supporting evidence from the candidate's CV and allows users to review extracted information before analysis. The system should also use verified sources for career trends and learning resources, clearly communicate AI limitations, and avoid presenting predictions as guaranteed outcomes. Users should review the recommendations and make their own decisions as they use generated results solely as a guide.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Charlotte]

[24/10/2026, Saturday]